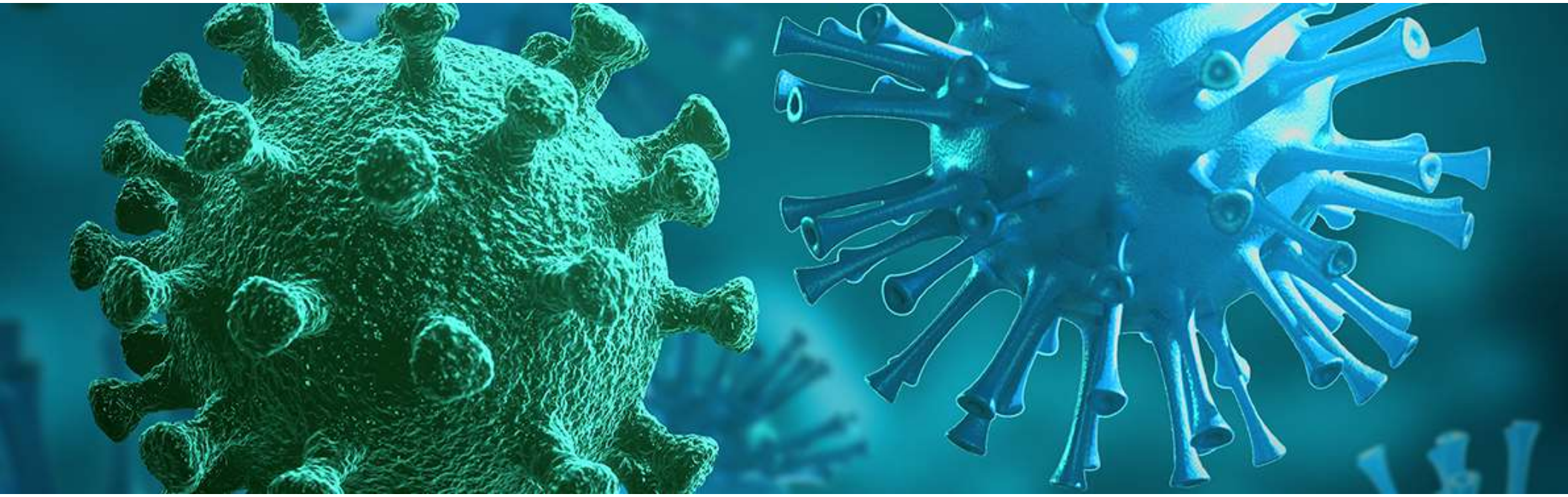
cobas® Influenza A/B & RSV UC on cobas® 6800/8800 Systems

Product Presentation

Presented by [Name, Surname, Function]



19 November, 2021 | © 2021 Roche | MC-US-10336

*cobas® Influenza A/B & RSV UC is a **research use only** product. **Not for use in diagnostic procedures.***

19 November, 2021 | © 2021 Roche | MC-US-10336

cobas® Influenza A/B & RSV UC

Sample type and assay specifics

Intended Use	Intended for simultaneous qualitative detection and differentiation of **influenza A virus, influenza B virus and respiratory syncytial virus (RSV) RNA in nasopharyngeal swab specimens** from patients with signs and symptoms of respiratory infection in conjunction with clinical and epidemiological risk factors to support the understanding of virus epidemiology for Population Health Management.
Sample types and collection media	Nasopharyngeal swab samples collected in • Copan Universal Transport Medium System (UTM-RT) or • BD™ Universal Viral Transport System (UVT) or equivalent
USAP	Assay available with ready-to-use USAPs enabling result call-out for: • Influenza A/B & RSV • RSV only
Test duration	Results are available within less than 3.5 hours after loading the sample on the system.
Kit size	192 tests (when used with the **cobas omni** Utility Channel kit)
Open-Kit stability	Up to 30 days stability of prepared kit, followed by 90 days stability after first usage
SW requirements	**cobas®** 6800/8800 System Software v1.4 or higher

*Source: **cobas®** Influenza A/B & RSV UC Test IFU RUO (09541250001-01EN)*

CONFIDENTIAL

cobas® Influenza A/B & RSV

Intended use
Assay Design
Performance

19 November, 2021 | © 2021 Roche | MC-US-10336

cobas® SARS-CoV-2 & Influenza A/B

Intended use for research use only

The **cobas®** Influenza A/B & RSV UC Qualitative Nucleic Acid test for use with the **cobas omni** Utility Channel on the **cobas®** 6800/8800 Systems is an automated, multiplex, real-time reverse transcription polymerase chain reaction (RT-PCR) assay for the timely in vitro **qualitative detection and discrimination of influenza A virus, influenza B virus, and respiratory syncytial virus (RSV) RNA in nasopharyngeal swab specimens from patients with signs and symptoms of respiratory infection in conjunction with clinical and epidemiological risk factors to support the understanding of virus epidemiology for Population Health Management.**

The **cobas®** Influenza A/B & RSV UC Qualitative Nucleic Acid test for use with the **cobas omni** Utility Channel on the **cobas®** 6800/8800 Systems **is intended for research use only and is not for use in diagnostic procedures.**

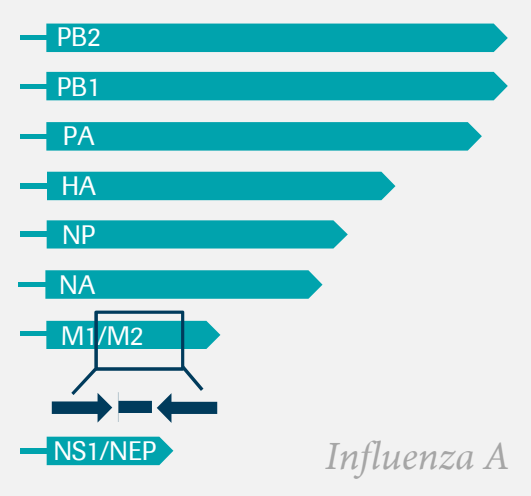
*Source: **cobas®** Influenza A/B & RSV UC Test IFU RUO (09541250001-01EN)*

CONFIDENTIAL

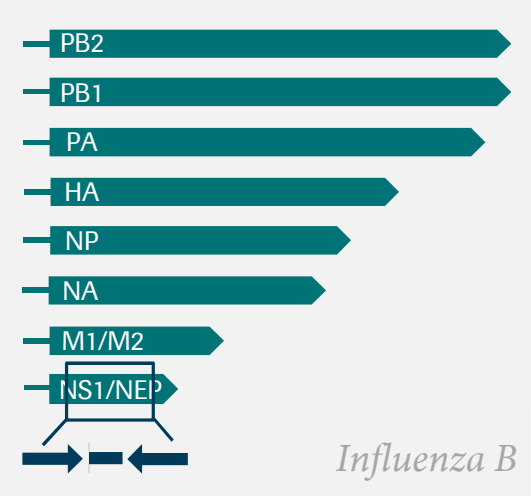
Assay Design for Influenza A and Influenza B

pan-Influenza A* and pan Influenza B assay

* Inclusive of G4 EA H1N1 swine virus with pandemic potential



PB2: PB2 protein gene
PB1: PB1 protein gene
PA: PA protein gene
HA: hemagglutinin protein gene
NP: Nucleoprotein gene
NA: Neuraminidase protein gene
M1: membrane protein 1
M2: membrane protein 2
NS1: nonstructural protein 1
NEP: nuclear export protein



The conserved region of Influenza A in segment 7 in the matrix protein (M1/M2) gene

A

The conserved region of Influenza B in segment 8 in the nuclear export protein (NEP) & nonstructural protein 1 (NS1) genes

B

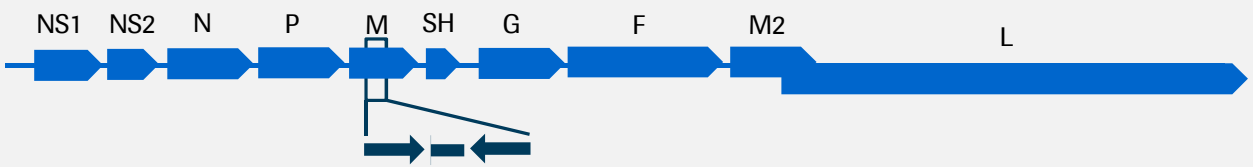
| 1. Honglei Suna, Yihong Xiaob, Jiyu Liua, Dayan Wang et al.. Prevalent Eurasian avian-like H1N1 swine influenza virus with 2009 pandemic viral genes facilitating human infection. 17204–17210, PNAS, July 21, 2020, vol. 117, no. 29
| 2. cobas* Influenza A/B & RSV UC Test IFU RUO (09541250001-01EN)

CONFIDENTIAL

Assay Design for RSV

pan-RSV assay, inclusive for RSV subtypes A and B

For RSV, selective amplification of target nucleic acid from the sample is achieved by the use of target-specific forward and reverse primers for the RSV matrix protein sequences.



NS1: nonstructural protein 1
NS2: nonstructural protein 2
N: Nucleoprotein
P: Phosphoprotein
M: Matrix protein
SH: small hydrophobic protein
G: Attachment protein
F: Fusion protein
M2: Regulatory protein
L: Large polymerase protein

| 1. Source genomic RSV structure: Hannah H Nam, Michael G Ison. Respiratory syncytial virus infection in adults. BMJ 2019;366:l5021 | doi: 10.1136/bmj.l5021 | 2. . cobas® Influenza A/B & RSV UC Test IFU RUO (09541250001-01EN)

CONFIDENTIAL

cobas® Influenza A/B & RSV UC

Transport and storage of specimens

Transport and storage of specimen for cobas® Influenza A/B & RSV UC

Media	condition
3 mL of Copan Universal Transport Medium UTM-RT® BD™ Universal Viral Transport (UVT), or equivalent.	stored for up to 48 hours at 2-25°C **followed by up to 3 days at 2-8°C and at ≤ -15°C for up to 30 days**
	Specimens are stable for **up to three freeze/thaw cycles when frozen in primary tubes at ≤ -15°C**

*Source: : **cobas®** Influenza A/B & RSV UC Test IFU RUO (09541250001-01EN)*

CONFIDENTIAL

cobas® Influenza A/B & RSV

Product concept
Workflow description
Result interpretation

19 November, 2021 | © 2021 Roche | MC-US-10336

Kit concept

cobas® Influenza A/B & RSV UC

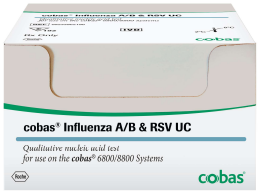
cobas omni Utility Channel kit

Number of tests	192
Stability once opened	90 days
Re-uses	40 runs

cobas® Buffer Negative Control Kit

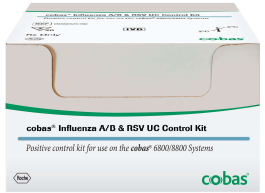
Number of negatives in kit	16 in single use format

cobas® Influenza A/B & RSV UC



Primer and probes	1 vial for 192 tests

cobas® Influenza A/B & RSV UC Control Kit



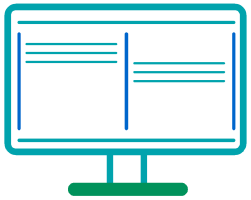
Positive Control	10 vials for 20 runs

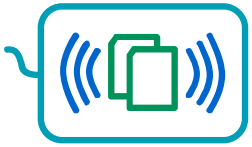
CONFIDENTIAL

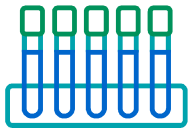
cobas omni Utility Channel Respiratory solution workflow

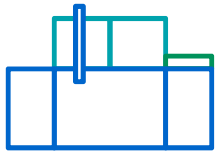
Simplified Utility Channel workflow – first use

Utility Channel software	RFID reader/writer	Reagent cassette	**cobas® 6800/8800** Systems







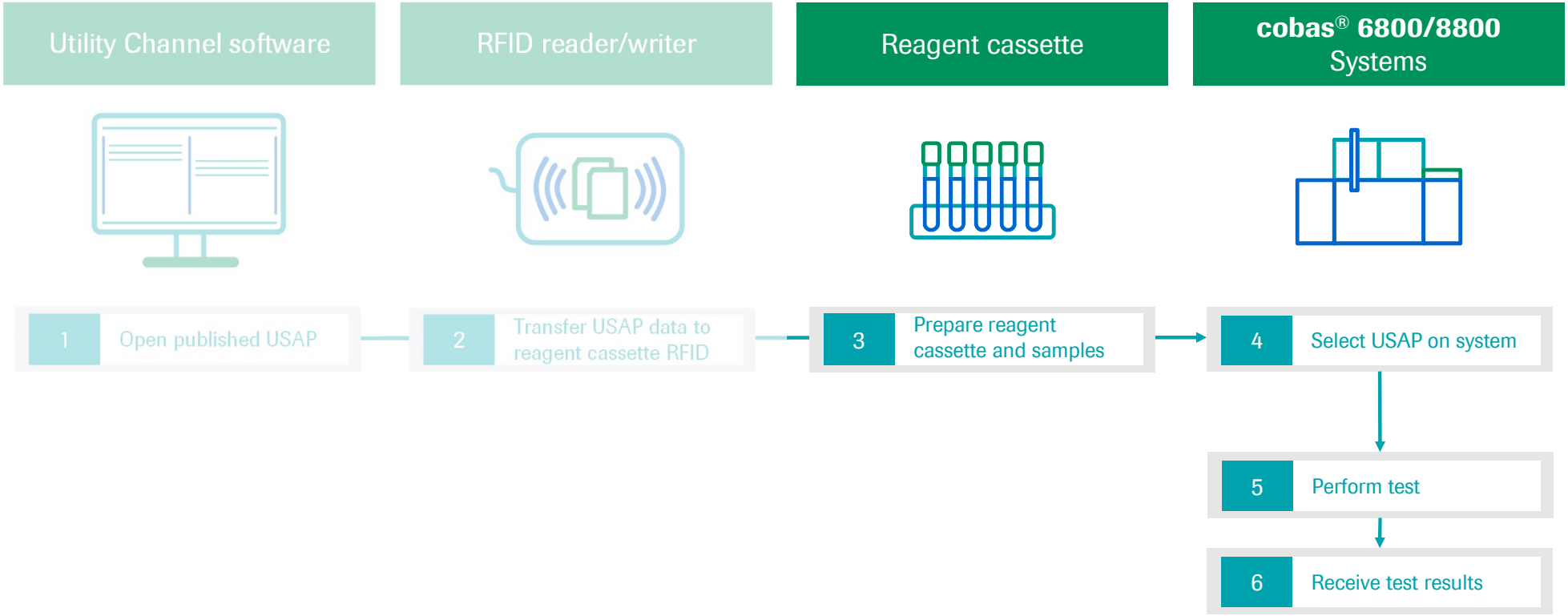


1. Install USAP profile on standalone PC
2. Transfer USAP data to reagent cassette RFID
3. Install USAP on system
4. Prepare reagent cassette and samples
5. Perform test
6. Receive test results

CONFIDENTIAL

cobas omni Utility Channel Respiratory solution workflow

Simplified Utility Channel workflow – routine use



CONFIDENTIAL

Quality control and validity of results

Validation of results is performed automatically by the **cobas**® 6800/8800 software based on negative control performance. The positive control validation has to be performed by the operator based on the positive control performance.

- One **cobas**® Buffer Negative Control [BUF (-) C] and one **cobas**® Influenza A/B & RSV UC Control [FluA/BRSV(+) C] need to be processed with each run.

Run and result validity interpretation

Validity	Control	Valid	Invalid	Validation
Run	BUF (-) C	Indicated as “Valid” in Test Result column	Indicated as “Invalid” in Test Result column (All samples of the run must be retested)	**cobas**® 6800/8800 Systems
Run	FluA/B-RSV (+) C	Ct value indicated in each Target column	Indicated as “Invalid” or “Negative” in one of the Target columns (2, 3 OR 4) (All samples of the run must be retested)	Operator
Sample Result	IC	Indicated as “Yes” in Valid column	Indicated as "No" in Valid column AND Target 2, 3 AND 4: Invalid (Invalidated sample must be retested)	**cobas**® 6800/8800 Systems

*Source: **cobas**® Influenza A/B & RSV UC Test IFU RUO (09541250001-01EN)*

CONFIDENTIAL

Result Interpretation

Target 2 (RSV)	Target 3 (Flu A)	Target 4 (Flu B)	Interpretation
RSV Negative	Any	Any	Target Result for RSV is valid. Result for RSV RNA is Not Detected.
RSV Ct value	Any	Any	Target Result for RSV is valid. Result for RNA A RNA is Detected.
Invalid	Any	Any	Target Result for RSV is invalid. Sample should be retested. If the result is still invalid, a new specimen should be obtained.
Any	**Flu A Negative**	Any	Target Result for Flu A is valid. Result for Flu A RNA is Not Detected.
Any	**Flu A Ct value**	Any	Target Results for Flu A is valid. Result for Flu A RNA is Detected.
Any	**Invalid**	Any	Target Result for Flu A is invalid. Sample should be retested. If the result is still invalid, a new specimen should be obtained.
Any	Any	**Flu B Negative**	Target Result for Flu B is valid. Result for Flu B RNA is Not Detected.
Any	Any	**Flu B Ct value**	Target Results for Flu B is valid. Result for Flu B RNA is Detected.
Any	Any	**Invalid**	Target Result for Flu B is invalid. Sample should be retested. If the result is still invalid, a new specimen should be obtained.
Invalid	**Invalid**	**Invalid**	None of the Targets Results are valid. Sample should be retested. If the result is still invalid, a new specimen should be obtained.

*Source: **cobas®** Influenza A/B & RSV UC Test IFU RUO (09541250001-01EN)*

CONFIDENTIAL

Q&A

19 November, 2021 | © 2021 Roche | MC-US-10336

Doing now what patients need next

19 November, 2021 | © 2021 Roche | MC-US-10336